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Bioprocess digital twins of mammalian cell culture for advanced biomanufacturing

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Today's biomanufacturing processes are still operated based on experience, and thus can hardly cope with increasing bioprocess complexity. Recently, there is a growing interest in industrial applications of the digital twins (DT) which integrate physical and virtual systems via real-time data monitoring, thus enabling their interactive communications for the enhanced operational efficiency towards advanced biomanufacturing. We suggest that bioprocess DT of cell cultures can be developed by incorporating in-line monitoring, advanced data analytics with machine and deep learning, and mechanistic models representing the mammalian cells and bioreactor for virtually mirroring their behaviors under adjustable process conditions. In this review, we summarize and highlight recent advances in the key components within bioprocess DT platform and discuss the current challenges with future research direction.

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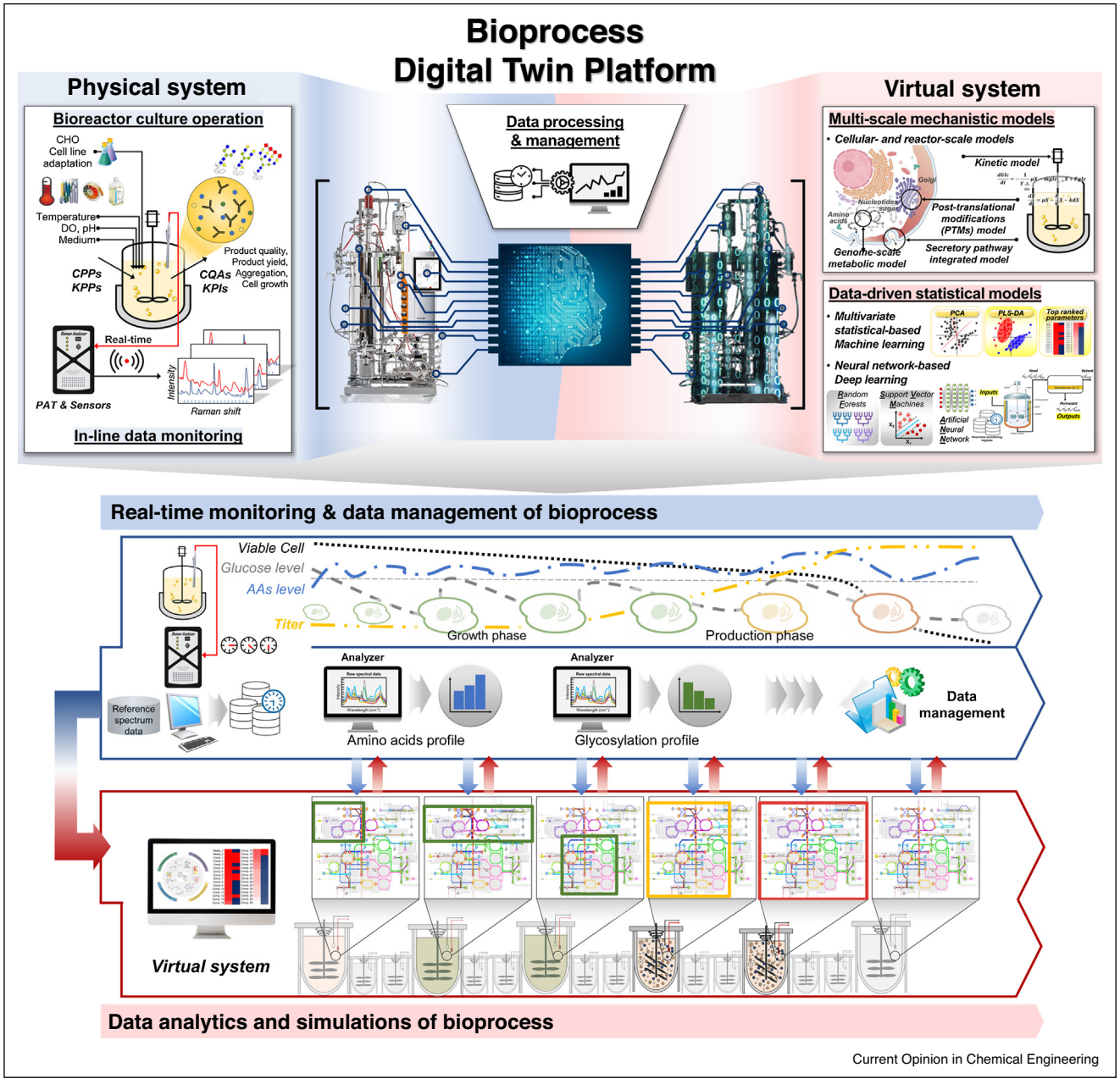
Introduction

Bioprocesses for producing a multitude of therapeutic proteins have operational difficulty and complexity since various process parameters and cellular traits significantly affect productivity and product quality during mammalian cell cultures, mainly by Chinese hamster ovary (CHO) cells which have been widely used as the workhorse for decades in biomanufacturing. For example, key performance indicators (KPIs) such as titer and yield as well as critical quality attributes (CQAs) are highly dependent of culture environments with adjustable key and critical process parameters (KPPs and CPPs) including media compositions and bioreactor conditions (e.g. pH, temperature, dissolved oxygen, etc.) [1]. In addition, process uncertainty and perturbation often lead to the increase in the product release time and total cost. Therefore, it is highly required to (1) closely monitor their dynamic behaviors, (2) better understand the relationships between KPPs/CPPs and KPIs/CQAs, and (3) finally identify relevant targets which can be timely manipulated to achieve enhanced culture performance, operational efficiency, and reliable product supply. They are indeed current challenges and trends towards next generation biomanufacturing, and can be realized by combining real-time monitoring and data management, statistical analysis, and mechanistic modeling of bioprocess and cellular behaviors within the 'interactive platform' between physical system and its digital replica, which is referred to as bioprocess digital twins (DT) [2].

As similarly applied to various manufacturing processes [3], bioprocess DT can incorporate the intertwined components including multiple physical operations of cell culture, real-time data monitoring of bioreactor, data processing and management, and mechanistic/statistical models of both cells and bioprocesses (Figure 1). For the real-time monitoring, the recent advances in sensor and process analytical technologies (PAT), for example, Raman and near infrared spectroscopies, allowed us to measure multiple process attributes and parameters in the bioreactor and collect a large sets of cell culture data. Such historical and real-time data can be processed and stored in database management system (DBMS) which connects physical and virtual parts. In the virtual system replicating the physical processes, that is, cells and bioreactor, their operational conditions and dynamic behaviors can be mechanically simulated at cellular- and reactor-scales by resorting to knowledge-based mathematical (*in silico*) models, while in parallel process inputs and outputs can be statistically correlated and analyzed by data-driven approaches such as machine learning (ML) and deep learning (DT) techniques. Thus, two complementary modeling approaches (mechanistic versus data-driven) can be synergistically combined to capture states of whole cell metabolisms/regulations and cultural behaviors, effectively identify process bottlenecks and key engineering targets, and suggest various operational strategies to manage the product quality, thereby improving

Figure 1



A concept of bioprocess digital twin platform.
Physical system, data processing and management, and virtual system constitute together a digital twin. The physical system consists of cell culture operation and in-line/real-time monitoring through different sensors. The virtual system consists of knowledge-based mechanistic and data-driven statistical models to mirror the physical system. The models are built on prior knowledge, historical data, and real-time collected data from the physical system to improve its robustness continuously. (CPPs: critical process parameters; KPPs: key process parameters; CQAs: critical quality attributes; KPIs: key performance indicators; PAT: process analytic technology; AAs: amino acids).

reliability, efficiency and efficacy of intricate bioprocess operations.

Although the presented DT platform is highly recommended for advanced biomanufacturing which is undergoing a paradigm shift of digitalization [4], there is no review on DT of mammalian cell culture from the bioprocess operational point of view. Thus, in this work, we aim to summarize recent achievements of each component of physical and virtual systems in the bioprocess DT of CHO cell culture and discuss the current challenges and limitations with future research direction.

Monitoring and data collection of physical bioprocesses

Toward in-line and real-time data collection of physical system

Process monitoring is one of key steps for DT platform development: relevant process data covering input conditions (KPPs and CPPs) and output performance (KPIs and CQAs) should be collected in-line and continuously, and digitally transformed in order to enable the communication between physical and virtual systems [5]. In-line analytical characterization techniques for bioprocess monitoring in real-time have evolved significantly in recent years within PAT paradigm [6], generating a large amount of data at different time intervals to guide process-wide automatic monitoring (Figure 2a). Analytically, Raman analyzer archives collected spectral data which are in turn preprocessed and converted into normal variates with relevant process values through chemometric methods for comparison of the in-line data and off-line measured parameters to characterize unknown process patterns [7]. For applications, various in-line Raman spectroscopy commercial sensors (e.g. Kaiser RXN2™, BioViserion™, inVia™ confocal, and ProCellics™ analyzers) have been successfully installed in micro-liter lab to commercial scale bioreactors for CHO cell culture with different operation modes to measure dynamic profiles of cells, proteins, and metabolites, and CPPs, together with CQAs/KPIs continuously without off-line sampling (Figure 2b). Initially, basic culture profiles for cells, nutrients, and by-products (such as glucose, glutamine, glutamate, lactate, ammonia) were measured in-line in pilot scale bioreactor, and then the scope of measuring process variables was expanded to cover amino acids [8], pH [9], osmolarity and cell viability [10] and antibody titer [11] in fed-batch bioreactor. In some applications, the monitoring systems were further utilized in adaptive feeding strategies to regulate feed rates and maintain relevant input conditions such as concentrations of glucose [12] and phenylalanine [13] within target ranges, as such giving rise to the improved culture performance. Interestingly, Eyster *et al.* [14] in-line monitored lactate and glucose, and maintained lactate concentration at 2 g/L by adjusting glucose and/or lactate feed, resulting in significantly reduced (68%) ammonium accumulation and more galactosylated glycoform of monoclonal antibody (mAb). Raman spectroscopy was also implemented for real-time monitoring of glycosylation site occupancy in mAb [15]. By identifying spectral regions which are relevant to presence or absence of oligosaccharide structures at the conserved asparagine sites of mAb (Asn-297 *N*-linked glycosylation) respectively, similar structures of target molecules can be distinguished at low concentrations even in complex and varying media compositions.

Advanced in-line Raman monitoring system and spectral data processing and analysis

For consistent and reliable collection of culture data, Raman-based monitoring system requires robust chemometric models which convert the spectral signals to corresponding physical values. However, chemometric models are mostly derived based on multivariate regression such as partial least squares (PLS), and specific to cell-lines or operation conditions (e.g. reactor scales and operation modes). Thus, increasing effort has been devoted over past years to develop more 'generic' chemometric models which are calibrated with multiple data sets of various physical conditions including different cell-lines, bioreactor scales, operation modes [16•,17••] and Raman device configurations [18]. Specifically, Santos *et al.* [16•] considered a wider range of culture conditions covering different CHO cell-lines, clones, media, reactor scales (2, 7, 15, and 10 000 L) and culture modes (fed-batch, perfusion) by proposing a systematic framework to evaluate each entity impact on the model performance. Recently, Tulsyan *et al.* [17••] proposed a ML based chemometric calibration framework, named just-in-time learning (JITL) method, which constructs a local model based on the renewed Raman spectral signals in real time instead of using the mixed dataset all at once.

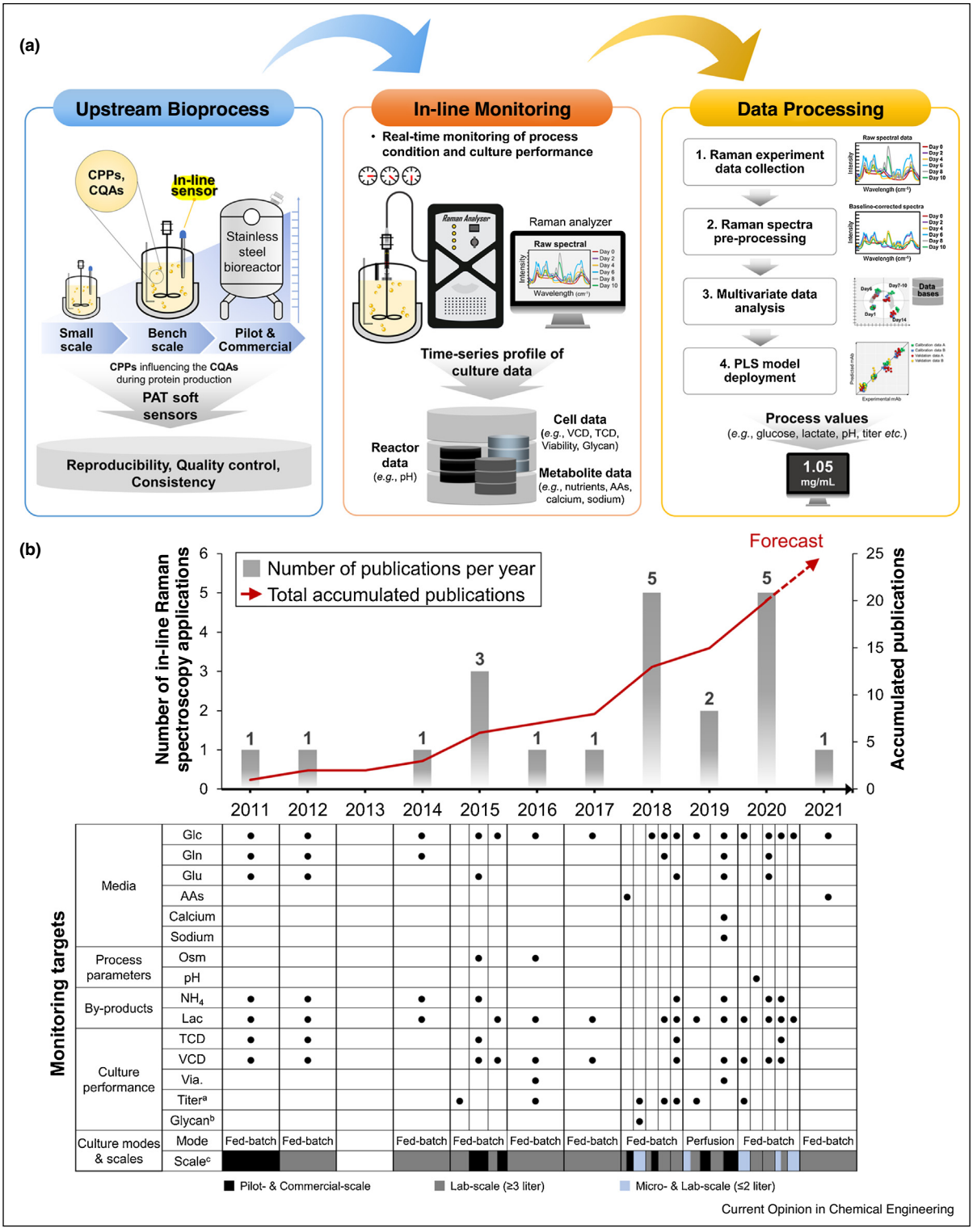
However, the use of Raman based monitoring techniques is still limited by the intrinsically weak signal as well as the strong fluorescence interference of several biomolecules in the culture broth [19], possibly causing adverse effects on the cells and bioprocess. Thus, deconvolution of Raman signal, combined with advanced analysis for complex biological compounds is indispensable to remove the distorted process information and thus enabling improved interpretation and deeper understanding of the process. In addition, various in-line biosensors should be developed to detect more quality-related process variables, such as aggregation, charge profile, and purity. Thus, in future, Raman spectroscopy can be combined with advanced data analysis skills and robust statistical models together with the advances in signal processing via ML and DL algorithms. Moreover, Raman spectroscopy can be integrated with other types of biosensors, for example, dielectric spectroscopy, to build multi-attribute sensing system which rapidly and simultaneously monitors the whole aspects of bioprocess and cell culture data [20], and ensures consistent quality of biotherapeutic products.

Multi-scale mechanistic models in virtual system

To apply DT in bioprocesses, digital system mirroring the physical processes virtually should be established by capturing the cellular characteristics and dynamic behaviors. To do so, we can use mechanistic models which have been developed at cellular-scales and reactor-scales.

From the cellular modeling perspective, the physiological behaviors and metabolic states of the mammalian cells under various environmental conditions can be elucidated by flux balance analysis (FBA) of genome scale

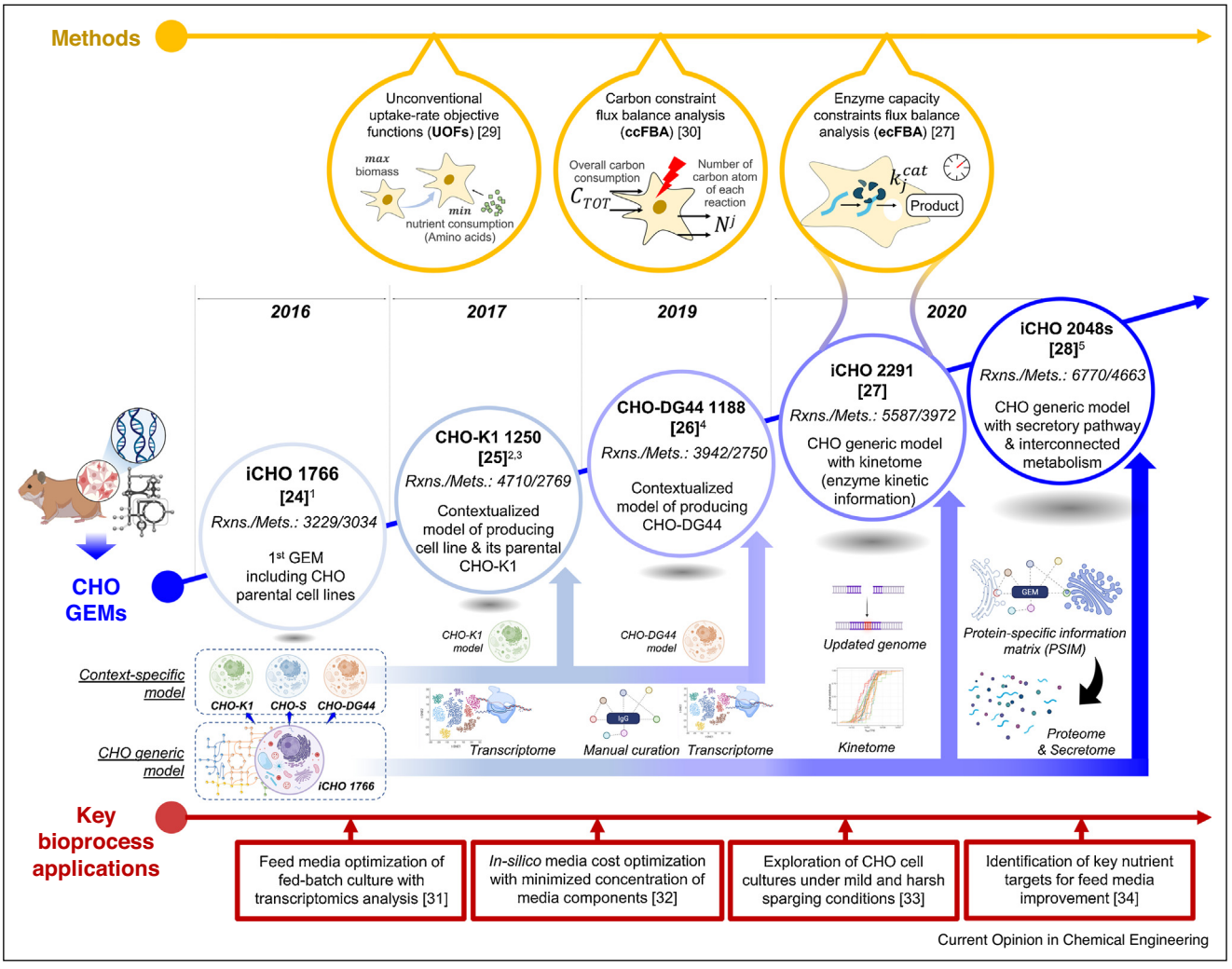
Figure 2



Monitoring of CHO bioprocesses using in-line Raman spectroscopy.
(a) A schematic of a general framework to construct Raman based bioprocess monitoring. **(b)** Timeline and development of Raman based in-line monitoring system in CHO cell culture bioprocesses as of April, 2021. In-line monitoring is starting to measure common metabolites and cell density since 2011 and is a growing interest in development and its application for monitoring product quality. (Glc: glucose; Gln: glutamine; Glu: glutamate; AAs: amino acids; Osm: osmolality; Lac: lactate; TCD: total cell density; VCD: viable cell density; Via: viability; [a]mAbs titer; [b]Measure glycosylation site occupancy; [c]Reactor scales from micro to commercial scale).

metabolic models (GEMs). Previously, mouse GEM [21] with CHO specific biomass equations was used to characterize CHO fed-batch cultures [22]. Since the first-ever CHO GEM [23] was successfully built based on CHO genome via world-wide community efforts [24], its quality has been consistently improved with context-specificity and extended coverage (Figure 3). A generic model of CHO cells (iCHO1766) was first reconstructed and then contextualized into parental cell lines such as CHO-K1, CHO-DG44 and CHO-S using transcriptome data [24], followed by characterizing antibody-producers derived from their parental cell lines (CHO-K1 and CHO-DG44) [25,26]. The model quality was also improved by network gap-filling, removal of duplicate/dead-end metabolites, and compartmentalization of the network, thus resulting in the increased numbers of genes, gene associated reactions and enzymes. For example, the recent CHO GEMs (iCHO2291 [27••] and iCHO2048s [28•]) have more than 2000 annotated genes with twofold and 1.5-fold increased reactions and metabolites compared to iCHO1766. In addition, iCHO2291 contains enzyme kinetic information with significant improvement on lipid metabolism, and *N*-glycan and *O*-glycan biosynthesis [27••] while iCHO2048s newly incorporated secretome and proteome information to describe secretory pathways related to CQAs. Interestingly, iCHO2048s predicts the increase in mAb secretion with gene knockdown approach (e.g. selectable marker *neoR*) [28•].

Figure 3



Timeline and development of genome-scale models (GEMs) for CHO cell lines.
Recent progression for GEM including various methods (yellow arrow line) and its applications (red arrow line) is represented with CHO GEMs as a major improvement. The central blue arrow shows recently updated models with the number of genes, reactions, and metabolites: (1),(3) represent that there are cell-line specific models such as DG44, S, SH87 and K1. Hypothetical name (e.g. CHO-SH1227, CHO-K1 1250 and CHO-DG44 1188) of reconstructed models with the number of their genes are shown in (2) and (4), and additional mammalian cell GEMs (e.g. human and mouse) in (5). (Rxns: reactions, Mets: metabolites).

For an accurate simulation of intracellular metabolism from the resultant flux distribution determined by FBA, several approaches have been exploited with intrinsic constraints and relevant cellular objectives [29,30]. For example, the lactate overflow metabolism in CHO cell culture was successfully deciphered by using enzyme capacity constrained FBA (ecFBA) of iCHO2291. Now, the model utility has been fully demonstrated through its industrial applications for feeding strategy [31], media cost improvement [32], and sparging strategy [33] during the process development. Another interesting application of GEMs was to identify key nutrient targets potentially promoting protein production and suggest their manipulations for feed media improvement [34].

Complementary to steady-state analysis by FBA, kinetic modeling is the most suitable approach for characterizing the dynamic behavior of cell growth and protein production upon limiting substrates and nutrients available during mammalian cell cultures in bioreactor. The kinetic models can be broadly classified into four types based on structured versus unstructured in the intracellular level, and segregated versus unsegregated considering the heterogeneity in cell population [35]. Since the models comprise a set of ordinary differential equations (ODEs) which are derived based on the mass balances of participating components using appropriate enzymatic kinetics and intracellular regulatory information, it is challenging to collect a large number of parameters even for a small model. Therefore, from bioprocessing point of view, it is highly desirable to develop multi-scale mechanistic models where various cellular processes related to the cell growth, protein production and product quality (metabolism, regulation, signaling, and glycosylation) are incorporated to clearly describe the links between CPPs and KPIs/CQAs at both cellular and reactor levels as exemplified elsewhere [36,37]. Recently, metabolic, regulatory and product quality related events are incorporated by unstructured and unsegregated kinetic model in conjunction with a small CHO metabolic and *N*-glycosylation networks, thus relating input process conditions and output product quality in bioreactor [38]. Similarly, dynamic changes in secreted mAb glycoform during CHO cell culture were elucidated through the integrative model coupled with an unsegregated and unstructured cell dynamics and glycosylation kinetics [39]. Thus, in this way, measurable CQAs such as glycosylation pattern, charge variants and aggregation by in-line sensors could be linked to the model outputs in bioprocess DT, as such allowing us to suggest feeding strategy [40] and adjustable key input variables [41] for controlling product titer and quality within the multi-scale model-based optimization framework.

Data-driven models and statistical analysis in virtual system

In parallel with mechanistic models, bioreactor data-driven virtual system within bioprocess DT platform comprises ML and DL models which statistically and analytically represent relationship between input conditions and output responses.

The ML models relating the cultural environments and physiological states of mammalian cells based on multivariate data analysis (MVDA) such as principle component analysis (PCA) and partial least square regression (PLS-R) can suggest key elements via the reduced high-dimensional culture data [42]. For example, Sokolov *et al.* [43] applied MVDA to correlate 20 input variables including media supplements and various process parameters with output CQAs and KPIs, thereby identifying potential key contributors for the enhanced culture performance. Similarly, effective process input conditions highly associated with product quality were selected by resorting to PCA, PLS-R, decision trees and genetic algorithm [44]. Recently, ML models were combined with Kalman filter for real time monitoring and control in CHO cell cultures to make them more reliable [45], and also integrated with a context-specific CHO GEM model to better predict dynamic concentration profiles of individual amino acid in culture medium [46]. In addition, phenotypic states of the CHO cell culture during different growth phases at lab to commercial scales were characterized by combining PCA, support vector machine, and GEM, finding enriched key metabolites for given phase [47].

However, the ML models may be insufficient to capture the complex and nonlinear relationship between CPPs and KPIs/CQAs due to the linear nature of MVDA. Thus, to overcome such inherent limitation of the MVDA, advanced data-driven models such as neural network-based DL with multi-layered structure, for example, artificial neural network (ANN), can be exploited when large amount of data are available. For example, dynamic concentration profiles of various nutrients including glucose, nitrate, and phosphate and cell growth data in algae-bacteria consortium were measured and used to train ANN model, followed by predicting the effect of nutrient consumptions on their growth with high accuracy [48•]. Similarly, ANN models were developed to simulate bioethanol production in *Saccharomyces cerevisiae* [49] and xylitol synthesis in *Debaryomyces nepalensis* [50] under a variety of process conditions (e.g. temperature, pH and agitation speed), and utilized to enhance the productivity. It should be noted that their applications to CHO cell culture are still in infancy due to the limited and sparse datasets available. However, with this increasing trend in data-driven modeling of bioprocess and advancements of data analytics techniques, we expect that a huge amount of mammalian cell culture data collected via in-line monitoring of bioreactor can be used for interactive communications across physical and virtual systems for the enhanced operational efficiency.

Conclusions

We presented bioprocess DT platform where physical and virtual systems are interactively connected via management system of collected culture data by in-line monitoring. In virtual system, multi-scale mechanistic and data-driven models are combined to mathematically and statistically relate input conditions (KPPs and CPPs) with output responses (KPIs and CQAs), respectively. In future, with a fully integrated and automated bioprocess DT, multiple CPPs and CQAs can be measured online via soft sensors, thus transforming the real-time culture profiles into virtual predictions of metabolic changes, cell growth, and recombinant product synthesis. Thus, it renders it possible to effectively identify moving targets to debottleneck the cell cultures dynamically and timely adjust them by advanced control strategies to improve culture performance, process efficiency, and reliable product supply towards advanced biomanufacturing.

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Declaration of Competing Interest

The authors report no declarations of interest.

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